



CALIBRATION REPORT

Report No.: ENV23CR460

Issue Date: 21-10-2024

Device ID: PM01P0013	Customer:
Product: Polludrone	Manufacturer: OIZOM
Validity: 1 year	

GAS CALIBRATION

Test Ambient conditions

Temp.: 20+/- 2 °C

Humidity: 54 +/- 10 %

Pressure: 14.56 psi

Test Operating conditions

Gas Temp.: 22+/- 2 °C

Gas Hum*.: 30+/- 10%RH

Gas Pressure: 2.94 psi

Gas Flow*: 2000 sccm

*using Alicat manufactured MFC sr nos. MCS-500SCCM-D, MC-500SCCM-D-SV with an accuracy of +/- 0.2 % FS | *using Nafion tube

CALIBRATION DATA (ppm)

Gas	Range	Resolu- tion	Zero Air Output	Std. gas conc.	Std. gas Output					Repeat- ability	RMSE
					1	2	3	Avg	Abs Error		
SO ₂	0-10	0.001	0	2.72	2.73	2.74	2.73	2.734	0.01	0.01	0.02
				5.4	5.39	5.37	5.38	5.38	0.02		
NO ₂	0-10	0.001	0	2.5	2.53	2.54	2.54	2.54	0.01	0.01	0.01
				5	5.04	5.01	5	5.01	0.01		
CO	0-5	0.01	0	2.67	2.68	2.69	2.78	2.68	0.01	0.01	0.02
				4	4.02	4.02	4.01	4.02	0.02		
O ₃	0-10	0.001	0	-	N/A	N/A	N/A	N/A	N/A	N/A	N/A

REFERENCE STANDARDS

(Note: CSLPL - Chemtron Science Laboratories Pvt; VGL - Vadilal Gases Limited)

SO₂: Certificate No. CSL-060224-13906-3-1 Make: CSLPL, Expiry: Feb. 2025;
NO₂: Certificate No. CSL-080524-10071-7-1 Make: CSLPL, Expiry: May. 2025;
CO: Certificate No. 13022024-4692 Make: VGL, Expiry: Feb. 2025;
Zero Air : Material No. FLAIR-G-S, Make: Inox Air Products.

Remarks: Repeatability of less than ±0.5% Full Scale was achieved for all parameters.

અહમદાબાદ વસ્ત્ર ઉદ્યોગ અનુસંધાન સંસ્થાન

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PM CALIBRATION

Parameter: PM_{2.5}, PM₁₀

Range: 0 - 5000 µg/m³

Resolution: 0.01 µg/m³

Reference: MetOne BAM-1020

Range: 0 - 1000 µg/m³

Resolution: 1µg/m³

Test Ambient conditions:

Temp Range.: 27- 34°C

Humidity Range: 63- 86 %

CALIBRATION DATA

Hour	PM _{2.5} (µg/m ³)		PM ₁₀ (µg/m ³)	
	Reference	Oizom	Reference	Oizom
1	39.20	47.84	188.87	190.60
2	31.21	37.49	169.29	179.39
3	26.41	33.56	190.30	231.32
4	24.44	34.34	152.56	181.65
5	31.96	35.05	136.65	175.38
6	27.35	32.80	148.27	189.46
7	40.70	41.27	155.82	187.71
8	45.31	46.28	161.43	174.73
9	58.47	53.22	138.41	168.43
10	54.80	52.43	170.34	183.71
11	39.86	43.92	171.97	182.29
12	39.57	42.74	172.48	186.66
Avg	38.27	41.74	163.03	185.94
Deviation	1.53 µg/m ³		7.91 µg/m ³	
Accuracy	96.00 %		95.14 %	

Remarks : Deviation meets the acceptable limit of >95%.

Oizom's Dustroid with OZPM_1 is certified against MCERTS performance standards for Indicative Ambient Particulate Monitors; Certificate No: CSA MC230419/00.

Tested by:		Approved by:	
Witnessed by:			

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COLLOCATION CALIBRATION REPORT

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Device ID: PM01P0013	Customer:
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COLLOCATION (with Reference CAAQMS)

Collocation time period: 48 hours

CO Collocation

Device ID	Average		MAE	MAPE (%)	RMSE	R ²
	Reference	Oizom				
CO (mg/m ³)	0.67	0.52	0.44	38.60	0.02	0.83
SO ₂ (ppb)	8.14	9.17	1.31	15.54	0.17	-
O ₃ (ppb)	21.08	20.69	5.12	21.87	0.57	0.90
NO ₂ (ppb)	9.08	9.23	3.99	60.51	0.29	0.87
PM _{2.5} (μg/m ³)	41.67	47.66	12.14	56.71	1.06	0.89
PM ₁₀ (μg/m ³)	113.08	116.60	24.02	12.62	1.99	0.91

REFERENCE

CO: Horiba APMA-370, Principle: Non-dispersive Infrared Analysis(NDIR)
NO_x: Horiba APNA-370, Principle: Chemiluminescence (CLD)
O₃: Horiba APOA-370, Principle: UV Absorption (NDUV)
SO₂: Horiba APSA-370, Principle: UV Florescence (UVF)
PM: The Met One Instruments, BAM-1020, Principle: Beta Attenuation Mass Monitoring

Remarks

Due to the low concentration range of SO₂ in the monitoring location, sensor readings do not have a linear relationship with the reference analyzer.



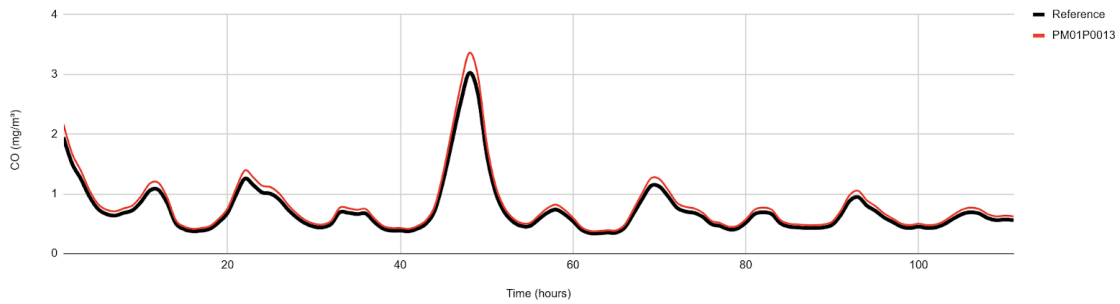
Tested by:		Approved by	
Witnessed by:			

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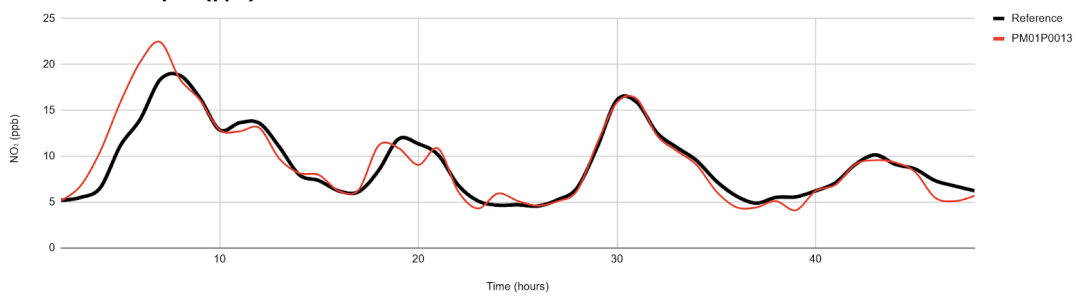
Annexure - A

COLLOCATION GRAPH

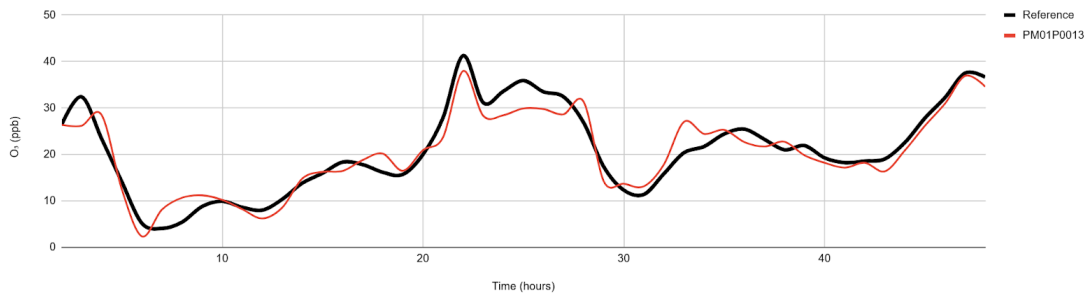
CO time series plot (mg/m³)



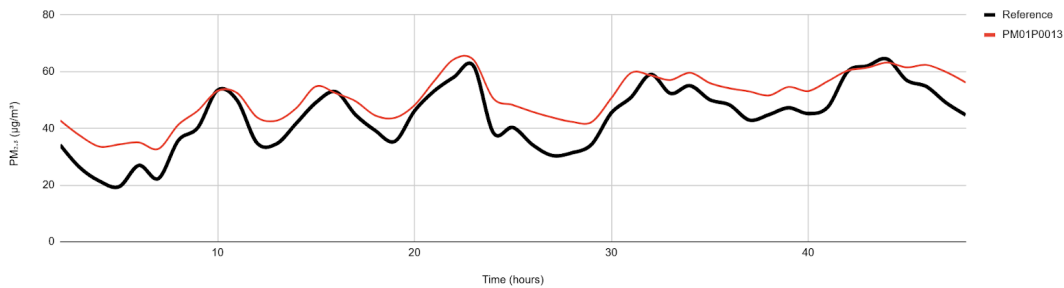
NO₂ time series plot (ppb)



O₃ time series plot (ppb)



PM_{2.5} time series plot (µg/m³)



PM₁₀ time series plot (µg/m³)

